AI251P Applied Computer Vision

Capstone Project – Phase 1 Report

Smart Pet Vision: Cat and Dog Classification

Smart Pet Vision · Phase 1 (M1–M3, 40 points)

Extended deadline: 29 October 2026, 11:59 PM

1 Team and project

Field	Details
Team name	Smart Pet Vision
Topic	Whole-image cat/dog classification; dated approval record not supplied.
Application area	Animal welfare and preparation of adoption-listing photographs
Code repository	`https://github.com/Yasser-20-alhajlah/smart-pet-vision`
W&B project	`https://wandb.ai/yasser-alhajlah-kaust/smart-pet-vision`
Dataset card	Appendix A; repository dataset card

Team members

Name	Student ID	Workstream	Phase 1 contribution
Abdulrahman Mastour Alharthi	225478	1. Data and classification	Data cleaning, split and audits; classifiers, evaluation, report and evidence
Yasser Mansour Alhajlah	225481	2. Detection	Derived body boxes and review; W&B; clean reproduction recorded on 23 October
Mohammed Adeeb Alrashdi	225488	3. Segmentation	Trimap masks and team-photo annotation guide
Musaad Obaied Saad Alqahtani	225471	4. Integration	Repository, configuration, W&B setup and reproduction instructions

Each member reviewed approximately 100 photographs in CVAT. Workstream labels follow Table 3 of the updated Capstone Project Guidelines.

Problem and acceptance criteria

- **Summary:** Support staff review of adoption photographs. Phase 1 establishes the data foundations and whole-image classifier; detection, segmentation, cut-outs and workflow integration belong to Phase 2.
- **Input → output:** A photograph → cat/dog label and class confidence; future detector crops feed the classifier.
- **Intended users and decision:** Shelter/adoption staff retain publishing authority. Low confidence, classifier/detector disagreement or no detected animal should trigger referral; 0.90 is a provisional confidence threshold.
- **Acceptance criteria (reported targets):** Test accuracy and macro-F1 targets are each ≥ 0.90. The selected model achieves 0.999096 and 0.998965, respectively. A dated record showing that these targets were fixed before evaluating test results is unavailable.

2 Data foundations (M1) (12 points)

2.1 Datasets

Dataset	Link / ref.	Content and annotations	Use	Licence	Images
Oxford-IIIT Pet	Oxford; Parkhi et al. (2012)	37 breeds; species/breed labels, trimaps and head boxes	Train / val. / test	CC BY-SA 4.0 (project record)	7,349
Team-collected (bonus)	Appendix A	401 photographs; 503 body boxes and masks	Previously evaluated external benchmark	Private; consent reported	401

The full Oxford dataset is used; no subset was selected before cleaning. The classification task has two outputs (cat and dog), not 37 breed outputs.

2.2 Cleaning

Check	Method / tool	Removed / retained
Corrupt / unreadable / small	Pillow open/verify; minimum side 32 pixels	0
Exact duplicates	MD5 of file bytes	0
Near-duplicates	pHash Hamming distance ≤ 4; union-find grouping	0; grouped
Other: empty trimaps	No usable foreground mask or body box	14
Images remaining	2,366 cats and 4,969 dogs; median 500×375 pixels	7,335

Near-duplicate audit: 45 pairs, 41 merged groups covering 84 images; largest group size 3.

2.3 Leakage-safe split

- **Grouping key:** Near-duplicate groups (pHash Hamming distance $\leq$ 4, merged transitively); 7,292 groups total.
- **Leakage controlled:** Near-identical photographs stay together. Zero duplicate groups and zero MD5 matches cross splits. The original code allocates groups before mask derivation and the 14 empty-trimap exclusions; counts below describe the retained population. This controls the documented duplicate relation; unavailable animal IDs and possible pretraining overlap prevent a claim of complete independence.
- **Stratification and seed:** Breed-stratified group allocation, targeting 70/15/15 by image count; seed 42; `data.py:group_stratified_split`.

Split	No. of groups	No. of images	Images per class
Train	5,091	5,123	1,652 cats / 3,471 dogs
Validation	1,103	1,106	357 cats / 749 dogs
Test	1,098	1,106	357 cats / 749 dogs

The smallest breed has 128 training images and 28 images in each evaluation split. Different photographs of the same pet may still cross splits; see Appendix B for diagnostic similarity filtering.

2.4 Label audit

Audited sample size	Errors found	Estimated label-error rate	Upper bound (95% confidence)
200 species labels	0	0.0%	1.5% (one-sided rule of three)
Same 200 breed labels	4 doubtful	2.0%	5.0%; Wilson CI 0.8–5.0%
300 derived body boxes	4 loose	1.3%	3.4%; Wilson CI 0.5–3.4%

Contact sheets were sampled using the project seed. Audit records label the initial review as AI-assisted and record team confirmation on 23 October 2026; reviewer and confirmation fields are retained. Zero observed species errors does not prove that every label is correct; the rule-of-three bound is approximate.

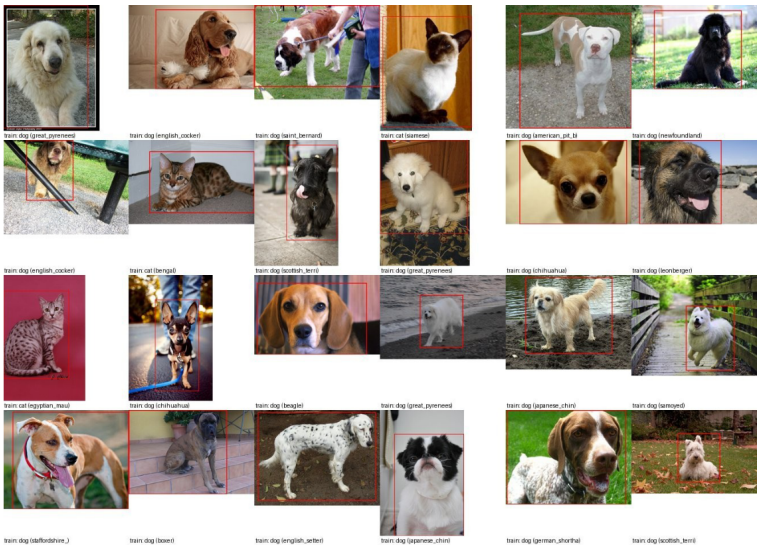


Figure 1: Body-box review examples. Data split: train; input resolution: native (variable); threshold: none. Boxes use trimap foreground and border, not head boxes.

2.5 Data loader and augmentation

Inputs: 224 × 224 pixels; ImageNet mean and standard deviation; batch size 64. The shuffled training loader uses torch.Generator with seed 42. Validation and test use resize to 255 px followed by a 224 px centre crop, without augmentation.

Training transform	Parameters	Application rationale
RandomResizedCrop	224 px; scale 0.5–1.0	Simulates loose framing and future detector crops
Horizontal flip	Probability 0.5	Orientation changes preserve species
ColorJitter	Brightness/contrast/saturation 0.2; hue 0.02	Models varied lighting while limiting coat-colour distortion

Breed stratification and macro-F1 address class imbalance in evaluation; no class weights or resampling are used. Training-mode dropout and augmentation mean training and validation metrics need not follow the same trajectory. Split roles are stored in runs/data/metadata.csv; private team images do not train these classifiers.

2.6 Dataset card summary

Oxford-IIIT Pet is dominated by well-lit web/pet-site photographs with one large, central animal. Dogs outnumber cats about 2:1; small, occluded, partly visible and multiple animals are underrepresented. Of 1,106 test body boxes, 1,105 are large. Shelter-domain performance therefore remains unestablished.

Zero species errors were found in a 200-image audit, but doubtful breed labels and loose derived boxes remain. Verified animal identities are unavailable. Possible ImageNet/Oxford overlap cannot be excluded, and similarity filtering cannot establish identity-disjointness or bound causal leakage.

The project records Oxford-IIIT Pet as CC BY-SA 4.0; images are downloaded from the official source and are not redistributed. Private team photographs are excluded from public releases. Collection records report consent, provenance, metadata removal and the exclusion of 13 suspect photographs; these claims have not undergone independent certification.

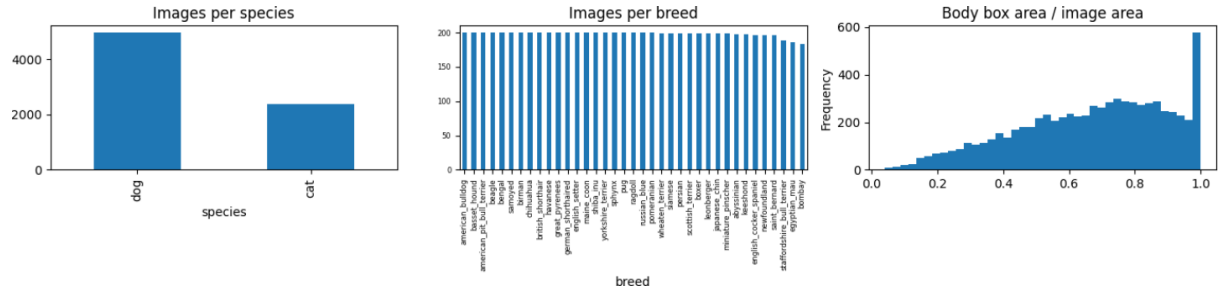


Figure 2: Cleaned dataset distribution: species, breeds and body-box area fraction. Data split: all 7,335 retained images, before splitting; input resolution: original/native (variable); decision threshold: none. Body boxes are derived from trimaps.

Private team collection

The team retained 401 of 414 collected photographs: 178 cat, 221 dog and 2 mixed-species images. There are 401 masks/label files and 503 body boxes matching the recorded animal counts. Four teammates corrected automatic annotation drafts in CVAT, including eight species labels.

This collection has already supported supplemental integrated-pipeline evaluation. Its Phase 2 metrics do not replace Phase 1 whole-image classification results. Future development needs a new independent final test after model and threshold decisions are fixed. The full dataset card is provided in Appendix A.

3 Classification (M2–M3) (20 points)

Task: whole-image binary classification of cat and dog; in Phase 2 the classifier is intended to verify detector crops within a staff-review workflow.

3.1 Runs compared

Checkpoints are selected by validation macro-F1, retaining the first epoch attaining the maximum. Test data do not select checkpoints or thresholds. Common recipe: AdamW, weight decay 1e-4, cosine schedule, batch size 64, mixed precision, cross-entropy and seed 42. Temperature is fitted on validation only.

Run (W&B link)	Backbone	Training	Input res.	Epochs	Val. acc.	Val. macro-F1
gh7ayx6h	Custom CNN	From scratch	224^2	25	0.927667	0.917516
xsz5afr9	ResNet-50	Pretrained, frozen	224^2	8	0.998192	0.997929
8nvxbsn2	ResNet-50	Pretrained, full fine-tuning	224^2	10	1.000000	1.000000

Strategy	Trainable parameters	Learning rate	Best epoch
Scratch	2,354,402	0.001	25
Frozen	4,098	0.001	6
Fine-tuned	23,512,130	0.0001	6

Scratch has five convolutional blocks with BatchNorm/ReLU/max-pooling, global average pooling, dropout 0.3 and a linear layer. Frozen training updates only a new 2048-to-2 head and freezes BatchNorm statistics. Full fine-tuning updates all ResNet parameters from ImageNet weights. Original runs use W&B group full-rerun-2026-10-18.

Different architectures, budgets and learning rates make this an application comparison, not an isolated causal ablation. Independent multi-seed variance has not been established.

Test-set results (final models only)

Run	Test accuracy (95% CI)	Test macro-F1	ECE (raw)
From scratch	0.905967 (0.887–0.922)	0.892300	0.013585
Pretrained, frozen	0.999096 (0.995–1.000)	0.998965	0.010967
Fine-tuned (selected)	0.999096 (0.995–1.000)	0.998965	0.001454

Test n = 1,106; 224 $\times$ 224 inputs; argmax; no decision threshold. Correct counts: 1,002 scratch and 1,105 for each transfer model. Accuracy confidence intervals are Wilson intervals; ECE uses 15 equal-width confidence bins.

Which model we keep and why:

Fully fine-tuned ResNet-50 has the highest validation macro-F1 (1.000 versus 0.997929 for frozen), meeting both Phase 1 test targets. Scratch misses the macro-F1 target. Exact McNemar p = 1.0 detects no difference between transfer models on this test set; it does not establish equivalence. Both differ from scratch (p < 1e-28). Frozen is a lower-cost alternative.

Fine-tuned epoch 6 validation accuracy is 100%; last epoch 10 validation accuracy is 99.9096%. These validation values must not be confused with test results.

3.2 Learning curves

Full-logging reruns use the same configurations, seed and data order. Shared training-loss, validation-accuracy and macro-F1 histories agree within 1e-12; runtimes differ. Training accuracy and validation loss below are rerun measurements, not recovered original logs. Original submitted checkpoints were retained.

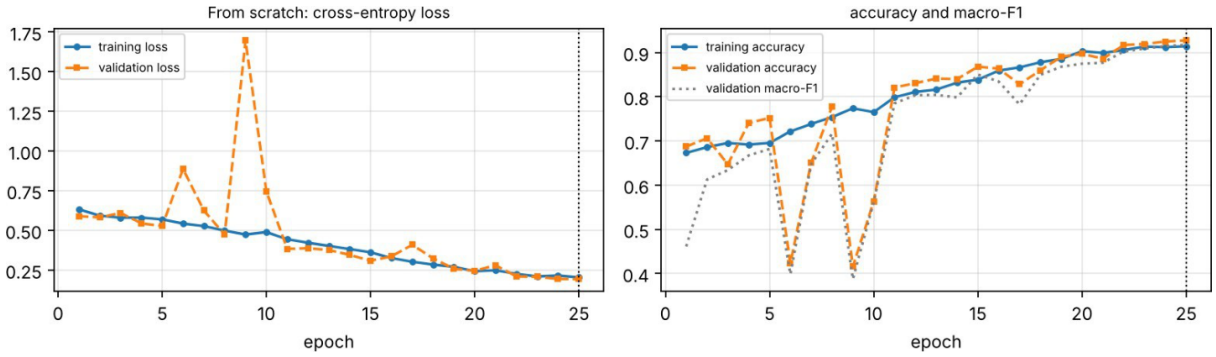


Figure 3: Scratch CNN training/validation loss and accuracy, plus validation macro-F1. Full-logging rerun bvns5vir. Data splits: train $n = 5{,}123$, validation $n = 1{,}106$; input resolution: 224×224; decision threshold: none (argmax).

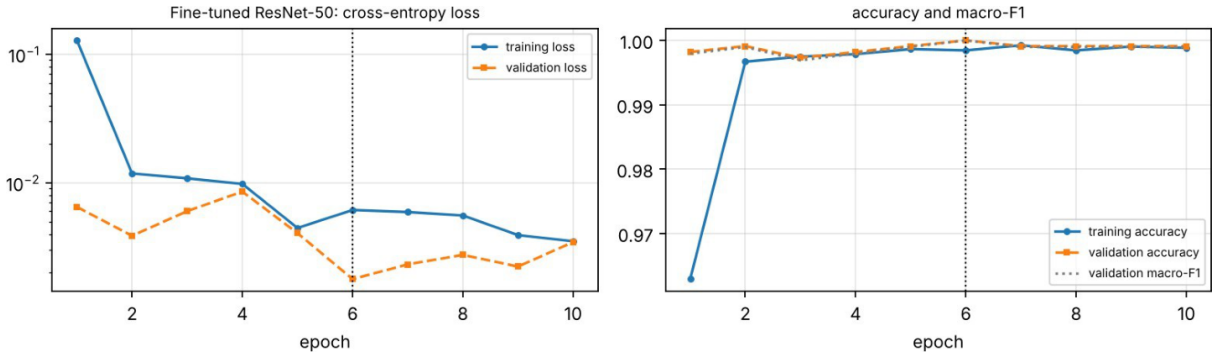


Figure 4: Fine-tuned ResNet-50 learning curves, with log loss axis. Rerun b3t0emrs. Data splits: train $n = 5{,}123$, validation $n = 1{,}106$; input resolution: 224×224; decision threshold: none (argmax). Dotted line: retained checkpoint. Training metrics average augmented batches.

Scratch appears under-trained: epoch 25 training accuracy is approximately 0.914 versus validation 0.928; losses are approximately 0.204 and 0.193. Augmentation/dropout contribute to the gap. Validation spikes may reflect evaluation-mode BatchNorm instability; this is an interpretation, not a measured causal effect.

Fine-tuned validation loss reaches approximately 0.0018 at epoch 6 and rises to 0.0035 by epoch 10, consistent with mild overfitting after the selected checkpoint. Frozen validation loss decreases from approximately 0.063 to 0.013; its full curve is recorded in rerun 4ixtdkiq.

3.3 Per-class results (selected model, test set)

Class	Precision	Recall	F1	Support
Cat	1.000000	0.997199	0.998597	357
Dog	0.998667	1.000000	0.999333	749
Macro average	0.999333	0.998599	0.998965	1,106

Fine-tuned model; test split; 224 × 224; argmax. Top-k accuracy (k = 2): 1.0; this is trivial for a binary classifier and is not informative.

3.4 Confusion matrix and calibration

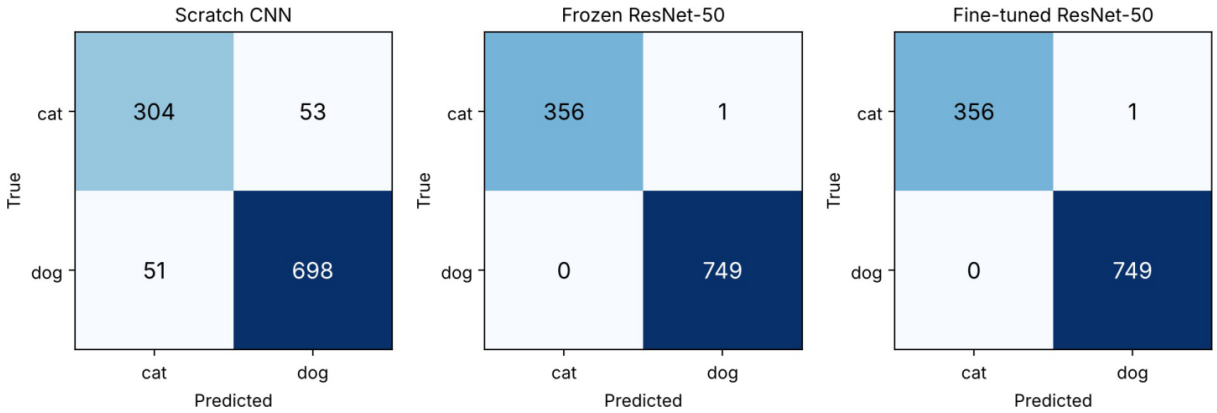


Figure 5: Test confusion matrices (rows: true; columns: predicted; cat/dog order). Scratch: [[304,53],[51,698]]; frozen and fine-tuned: [[356,1],[0,749]]. Data split: test $n = 1{,}106$; input resolution: 224 × 224; decision threshold: none (argmax).

Model	ECE raw	ECE calibrated	Saved temperature
Scratch	0.0135853811	0.0247622819	0.8230744600
Frozen	0.0109669426	0.0012126884	0.4895465076
Fine-tuned	0.0014543669	0.0009041591	0.05 (lower bound)

Temperature is fitted only on validation. Scaling worsens scratch ECE and lowers transfer-model ECE. Small ECE differences do not establish statistical significance or shelter-domain calibration; fine-tuned temperature reaches the optimization bound. The provisional 0.90 referral threshold is not certified for deployment.

The selected model makes one cat-to-dog error. Equal transfer-model accuracy and confusion matrices do not imply identical predictions: checkpoint inference on 28 October identified two image-level disagreements. Frozen misclassified a Sphynx cat; fine-tuning misclassified a Bombay cat.

Reliability diagrams

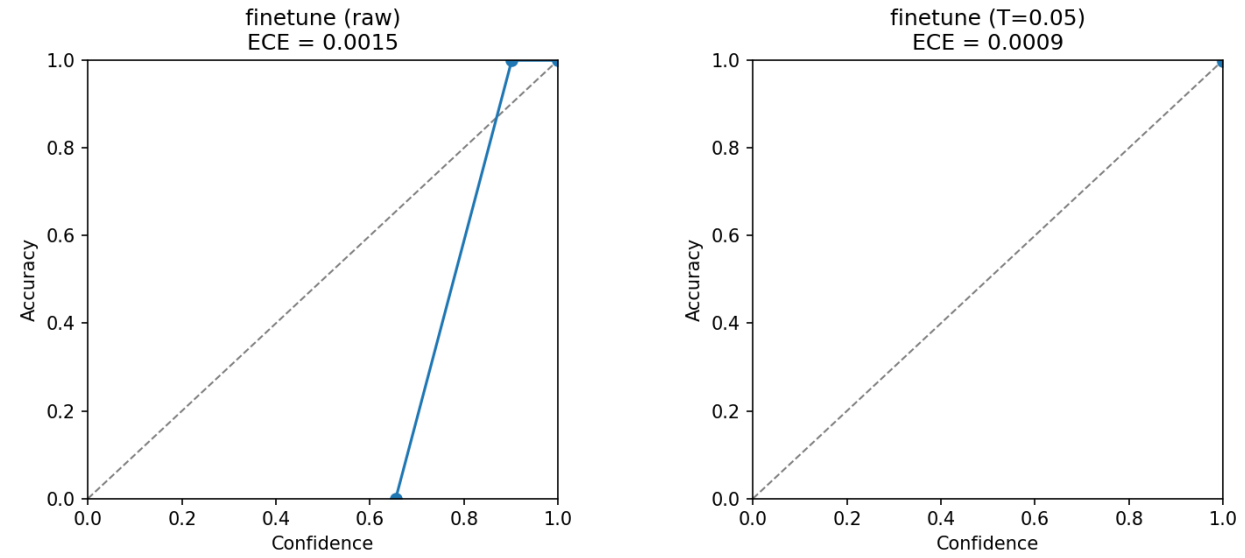


Figure 6: Selected fine-tuned model: raw (left) and temperature-scaled (right) reliability, $T = 0.05$. Data split: test $n = 1{,}106$; input resolution: 224×224; decision threshold: none (argmax). Fifteen equal-width confidence bins; temperature fitted on validation only.

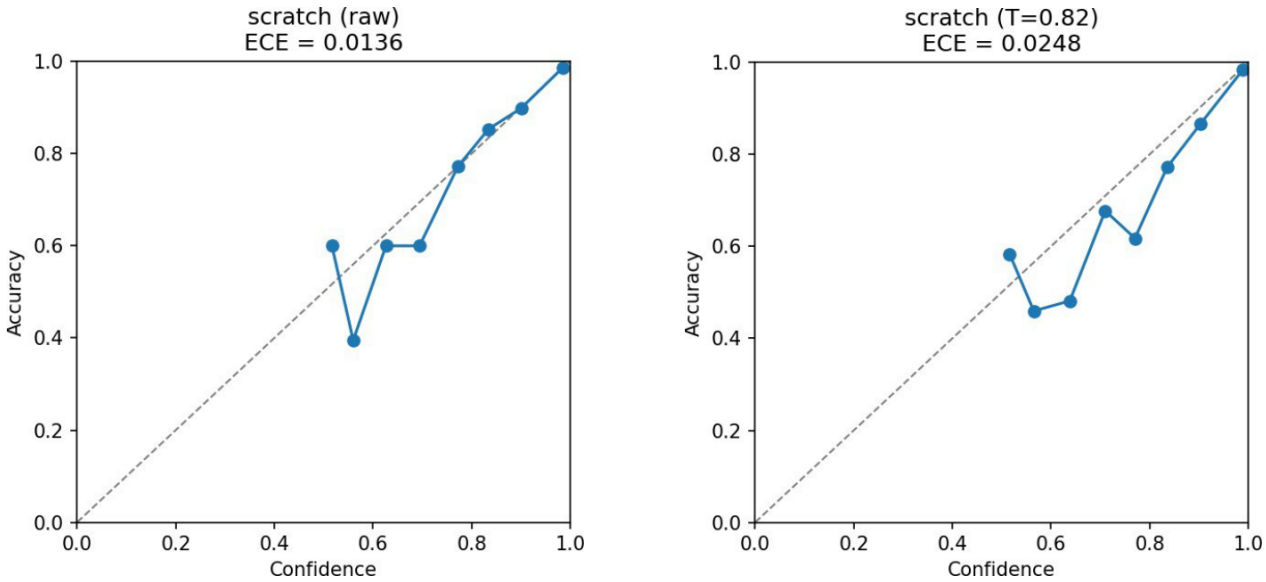


Figure 7: Scratch model: raw (left) and scaled (right) reliability, $T \approx 0.823$. Data split: test $n = 1{,}106$; input resolution: 224×224; decision threshold: none (argmax). Fifteen confidence bins; scaling worsens ECE.

Near-perfect validation separation limits interpretation of confidence calibration. The transfer models have high accuracy relative to their raw confidence on aggregate, while individual errors can still be overconfident. Lower aggregate ECE does not guarantee calibrated confidence for individual shelter photographs. Other raw/calibrated diagrams remain in the classification figures directory.

3.5 Error analysis

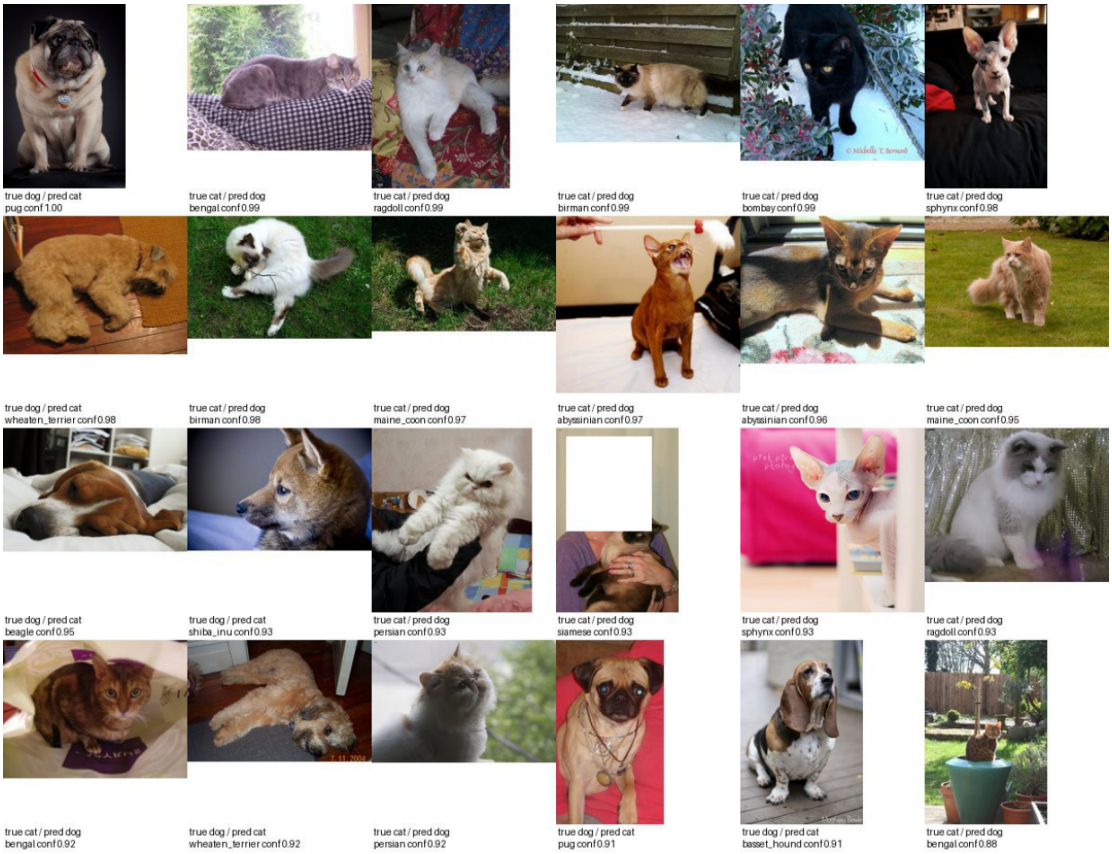


Figure 8: Most confident scratch-CNN errors; labels show true class, prediction and confidence. Data split: test; input resolution: 224 × 224; decision threshold: none (argmax). Visible face redacted for display only; model inputs and metrics are unchanged.

Error pattern	Examples	Likely cause / limitation
Long-haired / colour-point cats	Ragdoll 5; Birman 5; Persian 4; Maine Coon 4	Fur and shape cues may challenge the small CNN
Hairless / black cats	Scratch: Bombay 7/28; Sphynx 6/30	Appearance and contrast may contribute
Curled-up dogs	Pug 5; wheaten terrier 5; beagle 3; basset hound 3	Pose may hide shape cues
Label noise	0 species errors in the 200-image audit	Not excluded on individual images

Counts are breed-level test errors. Of 24 displayed scratch errors, 17 are cats predicted as dogs. Scratch makes 53 cat-to-dog and 51 dog-to-cat errors overall. Coat texture, low contrast and curled pose are gallery-based hypotheses, not established causal explanations.

3.6 Data-improvement plan

Finding from error analysis	Planned action	Expected effect / evaluation rule
Cat appearance errors; 2:1 dog imbalance	Collect black, hairless and colour-point cats; assess species rebalancing	Improve difficult-cat coverage; select changes on training/validation only
Small / occluded / unusual-pose animals are rare	Collect harder development photographs, including puppies and multiple animals	Improve domain coverage; use a new independent final test after decisions are fixed
Temperature hits lower bound	Compare calibration approaches on a harder separate calibration set	Assess confidence under domain shift; keep calibration and final-test roles separate
Four loose body boxes	Review trimap-derived boxes and add quality checks	Improve localization labels; evaluate detector changes in Phase 2

Expected effects are hypotheses to test, not measured gains. The 401-photo collection is a previously evaluated supplemental external benchmark; it cannot serve as a new untouched final test for subsequent development.

Limits of the evidence

Verified animal identities are unavailable, and possible ImageNet/Oxford overlap cannot be ruled out. Duplicate grouping is necessary but does not guarantee protection against every form of leakage. Diagnostic similarity filtering changes the evaluated population and cannot quantify a causal leakage effect (Appendix B).

There are no independent multi-seed runs establishing training variance. Architectures, learning rates and training budgets confound causal comparisons. High benchmark accuracy and small ECE do not establish reliable automatic publication for shelter photographs; staff retain publishing authority.

Collection consent and provenance are documented by the team, without external certification. A dated instructor-topic approval record was not supplied for independent verification. Evidence includes original training, full-logging reruns, checkpoint inference and saved-output verification; these address distinct aspects of reproducibility.

4 Reproducibility (8 points)

Item	Details
Repository and version roles	smart-pet-vision. Public source baseline inspected on 29 October: `875b275e9916965bb233b63783f8814ce3dab48a`. Release tag: `phase1-v1.0`. Publication snapshots do not establish the historical training commit.
Configuration file	`configs/config.yaml`, relative to the inner `smart-pet-vision/` repository.
Random seeds	42 for Python, NumPy, PyTorch and data loading.
Pinned requirements	`requirements.txt` pins training dependencies.
Reproduce the main result	In a separate Colab T4 environment, install `requirements.txt`; authenticate W&B; execute `notebooks/01_data_preparation.ipynb` then `notebooks/02_classification.ipynb`. Set `SPV_REPO` only if automatic discovery fails. This retrains and writes new results.
Saved-output verification	Recomputes metrics from saved predictions without inference or retraining. Run the command shown below from the inner repository.
Where data is obtained	Oxford images are downloaded at runtime from the official VGG source. Private team images are not required for Phase 1 training.
Reproduced by and result	The team records Yasser Alhajlah?s clean Colab T4 reproduction on 23 October 2026, with matching reported metrics. The archive and saved-output checks did not involve retraining. Separately, clean Colab T4 retraining was completed on 29 October 2026, as documented below.
Additional verification	28 October: float32 MPS inference using all three weights on 2,212 MD5-matching validation/test JPEGs. 29 October: public-archive SHA-256 checks and saved-output metric recomputation, without retraining.

Saved-output verification command:

```
python tools/verify_prediction_exports.py runs/verification_2026-10-28
```

Checkpoints, saved evidence and availability

Verified public assets: `phase1-v1.0` provides `Phase1_Checkpoints.zip` and `Phase1_Results_v2.zip`. Both retained archives match the current release SHA-256 digests and pass ZIP CRC checks. All 59 restored Results v2 files match their recorded hashes. Accuracy, macro-F1, confusion matrices and raw/calibrated ECE were independently recomputed from the three saved prediction exports. These checks do not retrain models or recover original GPU logits.

Release page | Checkpoints ZIP | Results ZIP

Restoration: Results v2 contains an enclosing directory. Copy its `data/`, `classification/` and `verification_2026-10-28/` directories directly under `smart-pet-vision/runs/`; do not retain the wrapper as another level. Work in a separate copy to protect submitted evidence.

Source identity: the public baseline above, the release-tag revision and older report-only revisions are publication records with different roles. The historical training commit is not established. The corrected source/evidence package is identified by `SOURCE_SHA256.txt`; original numeric evidence and weights are unchanged. Publication of the corrected package to GitHub remains pending.

Evidence scope: the Overleaf ZIP contains editable report text, tables, ten figures and verification notes. The separate source/evidence package supplies the corrected code, numeric supplements, image-sanitized historical notebooks and reproduction instructions. Weights and complete original result exports are available from the public release. Student-tested instructor access is recorded separately from artifact completeness and topic approval.

Clean retraining verification — 29 October 2026

All three classifiers were retrained and evaluated in a separate Google Colab T4 working directory using seed 42. The scratch, frozen and fine-tuned runs completed 25, 8 and 10 epochs, respectively. Checkpoints were selected using validation macro-F1.

Across all recorded epochs, the five logged learning-curve metrics matched the previous full-logging reference runs. The resulting test accuracy and macro-F1 also matched the reference results. This verifies same-seed reproducibility; it does not estimate variability across independent seeds.

The new outputs were saved separately from the original submission evidence.

Model	Run ID	Test accuracy	Test macro-F1
Scratch	`9fay48ne`	0.905967	0.892300
Frozen	`qlzja26v`	0.999096	0.998965
Fine-tuned	`2ptjc52f`	0.999096	0.998965

5 Submission checklist

The supplied `evidence/local_audit.json` records 30 passed local evidence checks (29 October 2026), including split separation, audits, checkpoint selection and saved-prediction metrics. These are saved-evidence checks, not fresh training or an independent access audit. Instructor access was tested by the student.

- ✓ Corrupt files and duplicates handled, with counts (Section 2.2).
- ✓ Near-duplicate groups stay within splits; identity limitations disclosed (Section 2.3).
- ✓ Label audit includes confidence bounds and reviewer provenance (Section 2.4).
- ✓ Dataset card covers licence, limitations and privacy (Section 2.6; Appendix A).
- ✓ Scratch, frozen and fine-tuned W&B runs linked (Section 3.1).
- ✓ Per-class metrics, calibration/ECE and both reported test targets verified (Section 3).
- ✓ Validation selects checkpoints; post-hoc test diagnostics are labelled (Appendix B).
- ✓ Data-improvement plan follows error analysis (Section 3.6).
- ✓ Fixed seed, pinned requirements and public version roles documented (Section 4).
- ✓ All figure captions state split, resolution and thresholds.
- ✓ Instructor access to submitted links confirmed by student testing (Section 4).
- ✓ Privacy review completed for ten bundled figures (faces, EXIF/GPS); external assets not audited.

6 Self-assessment

The self-assessment below uses the Phase 1 requirements in Table 6 of the updated Capstone Project Guidelines. Scores express the team's self-assessment, not instructor-awarded grades. No bonus is included.

Criterion	Points	Self-score	Strongest evidence	Verification limit
Data foundations and dataset card (M1)	12	12	Cleaning, grouped split, audit confidence bounds and limitations (Section 2; Appendix A)	Duplicate separation verified; animal identities and pretraining overlap remain limitations.
Classification (M2–M3)	20	20	Scratch/frozen/fine-tuned runs; per-class metrics, confusion, ECE and improvement plan (Section 3)	One seed; shelter-domain calibration remains unestablished.
Reproducibility	8	8	Config, fixed seed, 19 pinned dependencies, run IDs and Yasser's reproduction record (Section 4)	Public archive hashes and 59 restored files verified; historical training commit unknown; corrected source publication pending.
Total	**40**	**40**	**Team self-assessment**	Final grading remains with the instructor.

The team self-assessment is 40/40, not an instructor-awarded grade. Pre-results target timing, dated topic approval and historical training-source identity remain unverified. The ten bundled figures were reviewed; older external image assets are outside that privacy-review scope.

7 Obstacles and how they were addressed

Obstacle	Module	Impact on work	Action taken	Status
Head boxes only	M1	Body labels needed for later stages	Derived boxes from trimaps; reviewed 300; recorded 4 loose boxes	Addressed; residual error
14 empty trimaps	M1	No usable body annotations	Excluded; reported retained population	Addressed
Missing animal IDs	M1	Cannot verify animal-level independence	Grouped near-duplicates; reported diagnostic similarity analysis	Open limitation
Near-separable calibration data	M3	Temperature reaches lower bound	Disclosed bound; provisional threshold; proposed new calibration set	Open
Privacy / provenance	M1	Limits on sharing team photos	Private storage, consent records, metadata removal; excluded 13 photos	Evidence limits disclosed
Shared-folder overwrite risk	M1–M3	Could overwrite submitted evidence	Separate copies for verification and retraining	Controlled by workflow

Conclusion

Full fine-tuning achieves the strongest validation macro-F1 and meets both project targets on the held-out Oxford test split. Frozen transfer learning has the same test accuracy and macro-F1 at substantially lower training cost. The custom CNN performs less well and falls below the macro-F1 target.

Phase 2 will address detection and segmentation, harder development photographs and calibration under domain shift. New independent evaluation data are needed after development choices are fixed. Current benchmark results do not establish deployment performance on shelter photographs.

A Dataset card

Field	Dataset record
Dataset and task	Oxford-IIIT Pet (Parkhi et al., 2012); whole-image cat/dog classification. 37 breeds: 12 cat and 25 dog breeds.
Licence / download	CC BY-SA 4.0 as recorded by the project; official VGG source. Original image files are not redistributed.
Population	7,349 source images; 14 empty-trimap exclusions; 7,335 retained. 2,366 cats and 4,969 dogs; median 500×375 pixels.
Cleaning	Pillow verification; minimum side 32 pixels; MD5 exact checks; pHash Hamming $\leq$ 4 and union-find groups. 45 pairs, 41 merged groups, 84 images.
Split	Seed 42; group-aware, breed-stratified 70/15/15. Train 5,123 (1,652/3,471 cats/dogs); validation 1,106 (357/749); test 1,106 (357/749). Zero group/MD5 crossings.
Annotations	Body foreground = trimap foreground plus border. Body box = tight rectangle around this mask. Border ambiguity matters for later segmentation evaluation.
Audits	200 species labels: 0 errors; approximate 95% upper bound 1.5%. Four doubtful breed labels: Wilson CI 0.8–5.0%. Four loose boxes among 300: CI 0.5–3.4%.
Bias / limits	Large central pets, mostly well-lit web photos; dogs about 2:1. Small, occluded and multiple animals are rare. Missing IDs and possible pretraining overlap limit independence claims.
Sensitivity study	Nearest same-breed ImageNet-feature matches filtered diagnostically; retained-subset performance cannot bound causal leakage or establish identity-disjointness.
Intended use	Academic classification comparison and later staff-review support; no automatic publication authorization established.
Storage	Public metadata, audit exports, model outputs and hashes are separate from original Oxford files and private team photos.

A.1 Team-collected dataset

Of 414 received photographs, 401 were retained after excluding 13. Records distinguish 104 owner photographs received as screenshots and 297 supplied by team members; 170 of the latter retained camera EXIF before metadata removal. These categories do not establish who personally photographed every image.

Audit field	Recorded value
Images / masks / labels	401 / 401 / 401
Species	178 cat; 221 dog; 2 mixed
Body boxes	503; counts match n_animals
Readable / aligned	All photographs readable; masks match image dimensions
Dimensions / metadata	All photographs at most 2048 pixels; collection audit records EXIF removal
Duplicates	No exact duplicates; no reported MD5/pHash match with Oxford
Annotation	Automatic drafts corrected by four teammates in CVAT; eight species-label corrections
Mask share	Median approximately 0.28
Consent	Documented in team collection records; no independent certification

Stored privately under SmartPetVision_TeamData, excluded from public release packages. The public repository and W&B report contain instructor-facing project evidence. The collection has already supported integrated-pipeline evaluation as supplemental Phase 2 evidence, so it is not an independent previously unused final test for later development.

Phase 2 context (outside this assessment)

Team-stated reference targets: mAP50 $\geq$ 0.75, recall $\geq$ 0.80, mIoU $\geq$ 0.75, Dice $\geq$ 0.85, service success $\geq$ 99%, p95 latency $\leq$ 2 seconds on T4 and ONNX parity $\leq$ 0.001. These are targets, not certified operating performance.

B Image-similarity sensitivity analysis

Each test image was paired with its nearest same-breed training image using ImageNet ResNet-50 features: median similarity 0.80, 99th percentile 0.955, maximum 0.98. Close matches were filtered and accuracy recomputed; similarity is a proxy and does not verify animal identity.

Test images kept	Removed	n	Fine-tuned acc. (95% CI)	Scratch acc. (95% CI)
All	0	1,106	0.999 (0.995–1.000)	0.906 (0.887–0.922)
Similarity < 0.95	22 (2.0%)	1,084	0.999 (0.995–1.000)	0.905 (0.886–0.921)
Similarity < 0.90	133 (12.0%)	973	0.999 (0.994–1.000)	0.902 (0.882–0.919)
Similarity < 0.85	308 (27.8%)	798	0.999 (0.993–1.000)	0.892 (0.869–0.912)
Similarity < 0.80	562 (50.8%)	544	0.998 (0.990–1.000)	0.884 (0.855–0.908)
Similarity < 0.75	790 (71.4%)	316	0.997 (0.982–0.999)	0.851 (0.808–0.886)

Test split; 224 × 224 inputs; argmax; Wilson intervals. The listed similarity cutoffs select subsets, not classifier decisions. Fine-tuned accuracy is 1105/1106 = 0.999096 on the full set and 315/316 = 0.996835 at the strictest filter. The difference is 0.226 percentage points across different populations, not a causal leakage estimate.

Filtering changes test difficulty, breed mix and class composition. This experiment cannot prove absence of identity leakage, quantify its causal effect or provide an upper bound on leakage. Animal independence remains unverified; ImageNet pretraining overlap remains unexcluded.

These subsets were examined after final test results were available. They are post-hoc diagnostics, not additional independent final tests or a basis for selecting checkpoints or thresholds.

References and experiment provenance

1. Parkhi, O. M., Vedaldi, A., Zisserman, A., and Jawahar, C. V. (2012). *Cats and Dogs*. IEEE Conference on Computer Vision and Pattern Recognition. Dataset: https://www.robots.ox.ac.uk/~vgg/data/pets/.

2. Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). *On Calibration of Modern Neural Networks*. ICML. https://proceedings.mlr.press/v70/guo17a.html.

Original W&B runs: scratch gh7ayx6h, frozen xsz5afr9, fine-tuned 8nvxbsn2. Full-logging reruns: scratch bvns5vir, frozen 4ixtdkiq, fine-tuned b3t0emrs. Checkpoint-verification exports are recorded in the public results archive. These records distinguish original training, full-logging reruns and subsequent checkpoint verification.